

Title: Basic router configuration and static routing in Packet Tracer**THEORY:****Router**

A router is a network device that directs data packets between different computer networks. It manages traffic on the Internet, forwarding packets from one router to another across interconnected networks until they reach their destination.

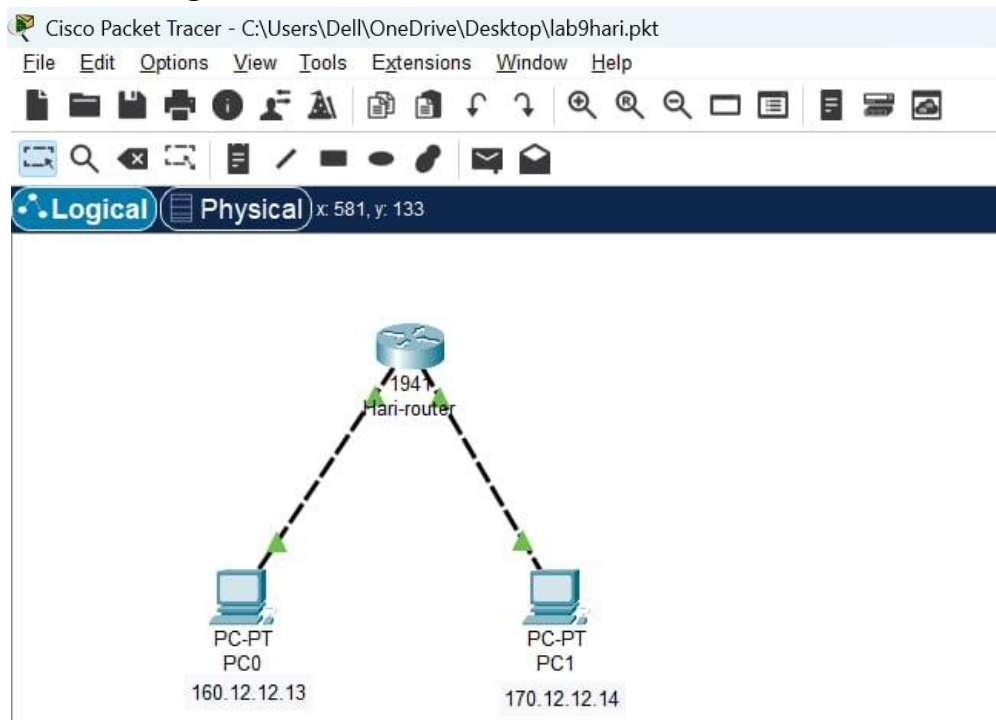
Network Diagram

Fig: Network diagram including router and switch

IMPLEMENTATION SEQUENCE

Here is the implementation sequence for Basic router configuration and static routing in Packet Tracer.

a. Basic router Configuration**i. Configuring Global Parameters**

Here are the steps to configure global parameters on a router:

Step 1: Open Packet Tracer and set up the devices (routers, PCs).

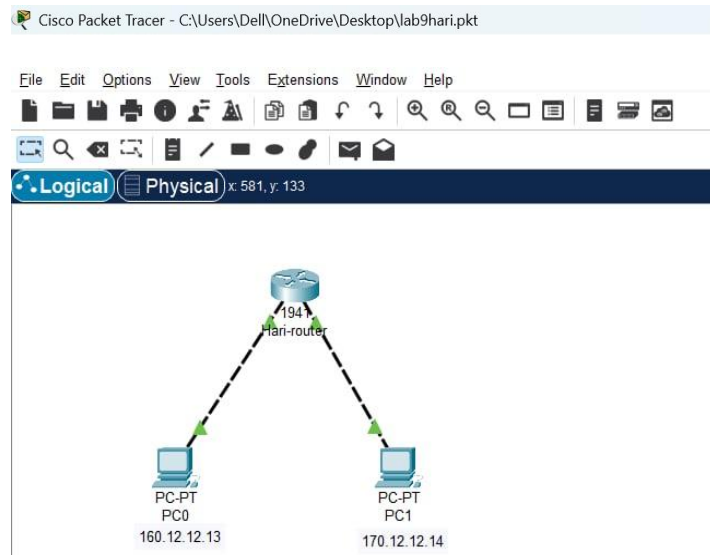


Fig: Simple Network setup

Step 2: Access the Router CLI

Use the console cable to access the router's command-line interface (CLI)

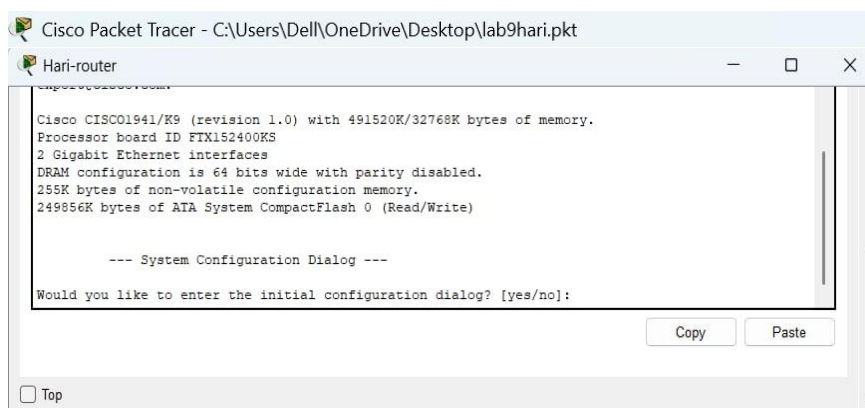


Fig: Accessing Router Command-Line Interface via Console Cable

Step 3: Configure Terminal

Type enable to enter privileged EXEC mode and use configure terminal to access global configuration mode.

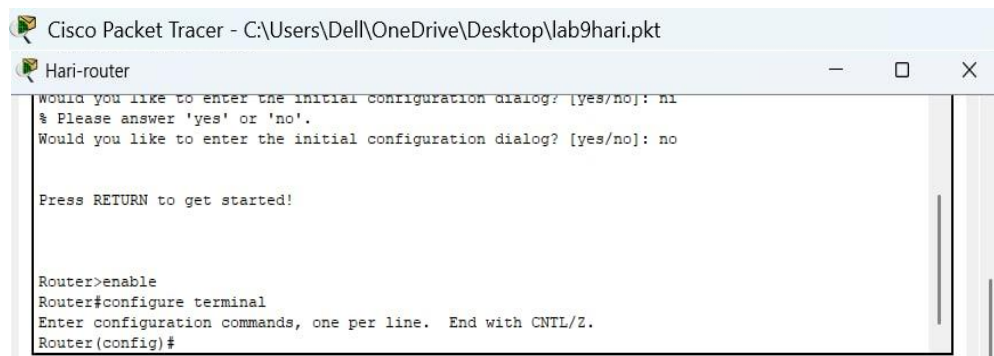


Fig: Entering Privileged EXEC and Global Configuration Mode on the Router

Step 4: Set a hostname for the router

Use the command `hostname [RouterName]` to assign a name to the router.

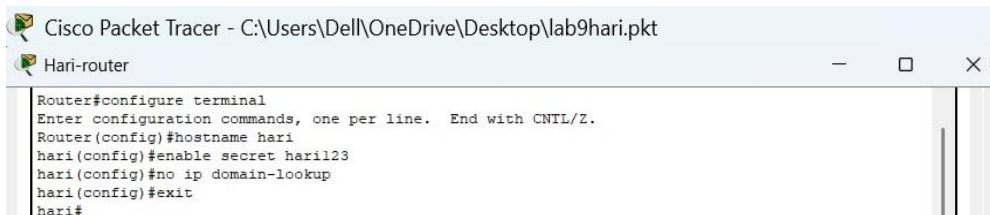


```
Cisco Packet Tracer - C:\Users\Dell\OneDrive\Desktop\lab9hari.pkt
Hari-router
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname hari
```

Fig: Setting the Hostname on the Router

Step 5: Configure a Password for Privileged Mode

Set an enable password using `Router(config)#enable secret [password]`



```
Cisco Packet Tracer - C:\Users\Dell\OneDrive\Desktop\lab9hari.pkt
Hari-router
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname hari
hari(config)#enable secret hari123
hari(config)#no ip domain-lookup
hari(config)#exit
hari#
```

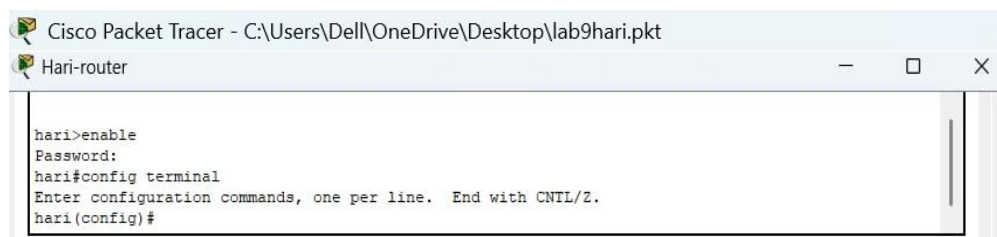
Fig: Configuring a Password

ii. Configuring Gigabit Ethernet

Here are the steps to configure gigabit Ethernet interfaces on a router:

Step 1: Configure Terminal

Type `enable` to enter privileged EXEC mode and use `configure terminal` to access global configuration mode.

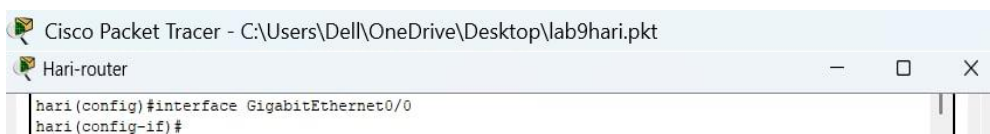


```
Cisco Packet Tracer - C:\Users\Dell\OneDrive\Desktop\lab9hari.pkt
Hari-router
hari>enable
Password:
hari#config terminal
Enter configuration commands, one per line. End with CNTL/Z.
hari(config)#
```

Fig: Entering Privileged EXEC and Global Configuration Mode on Router

Step 2: Access the interface configuration mode

Use the command `interface <interface_name>` to enter the configuration mode for a specific router interface.

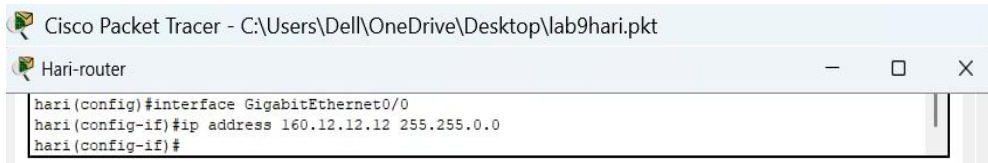


```
Cisco Packet Tracer - C:\Users\Dell\OneDrive\Desktop\lab9hari.pkt
Hari-router
hari(config)#interface GigabitEthernet0/0
hari(config-if)#
```

Fig: Accessing the interface configuration mode

Step 3: Assign an IP address

Once in the interface configuration mode, assign a unique IP address and subnet mask to the interface using the Ip address command.

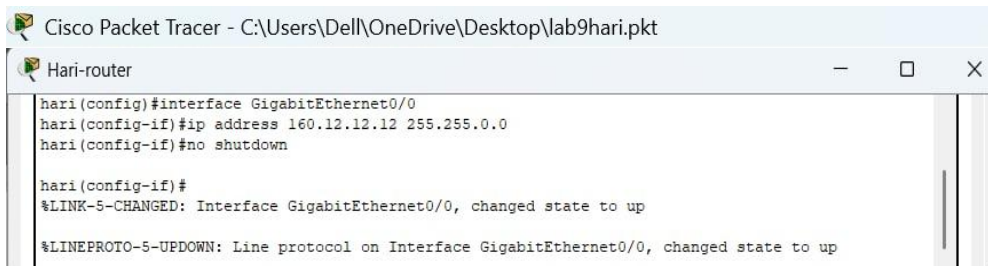


```
Cisco Packet Tracer - C:\Users\Dell\OneDrive\Desktop\lab9hari.pkt
Hari-router
hari(config)#interface GigabitEthernet0/0
hari(config-if)#ip address 160.12.12.12 255.255.0.0
hari(config-if)#
```

Fig: Enabling the Gigabit Ethernet Interface on Router

Step 4: Enable the interface

After assigning an IP address, use the no shutdown command to activate the interface.



```
Cisco Packet Tracer - C:\Users\Dell\OneDrive\Desktop\lab9hari.pkt
Hari-router
hari(config)#interface GigabitEthernet0/0
hari(config-if)#ip address 160.12.12.12 255.255.0.0
hari(config-if)#no shutdown

hari(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
```

Fig: Enabling the Gigabit Ethernet Interface on a Router

b. Static Routing Configuration

I.Configuring Network (PCs and Routers)

Network Diagram

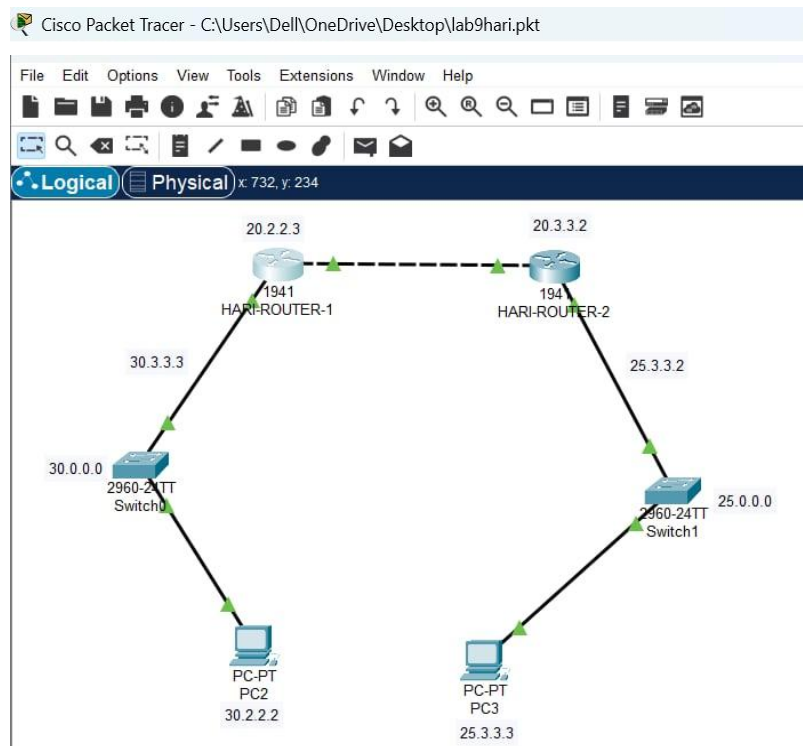


Fig: Network Diagram for static routing

-Hari Bashyal

Configuring PCs

Step 1: Assign IP addresses to the PC's

Configure the IP addresses on the PCs within the same network as their respective router's interface.

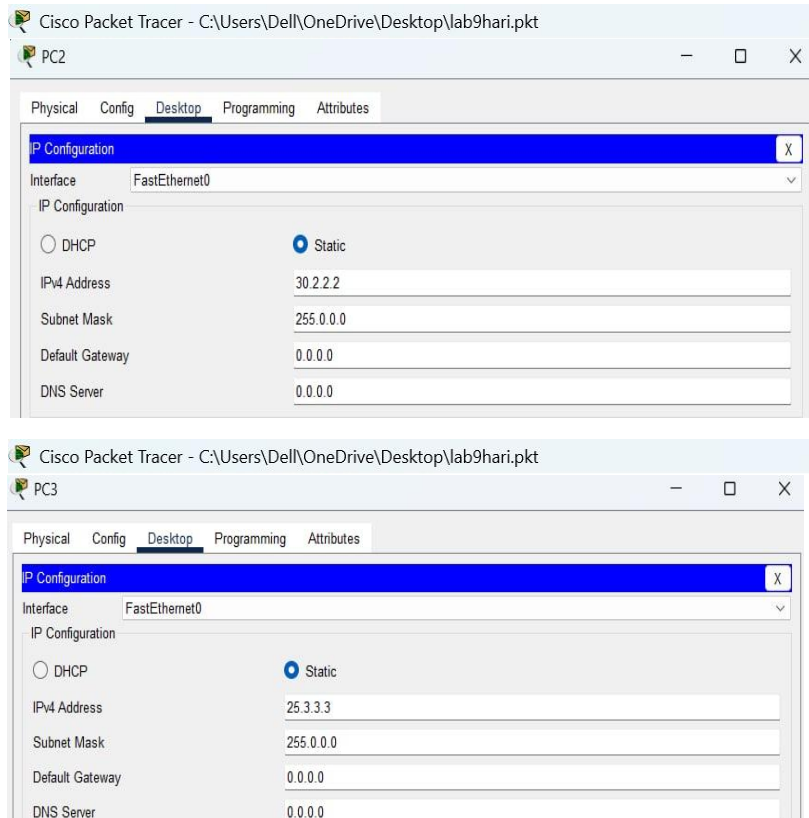


Fig: Assigning IP address to PC's

Configuring Routers

Step 1: Set the IP address on the Gigabit Ethernet interface

For HARI_ROUTER-1:

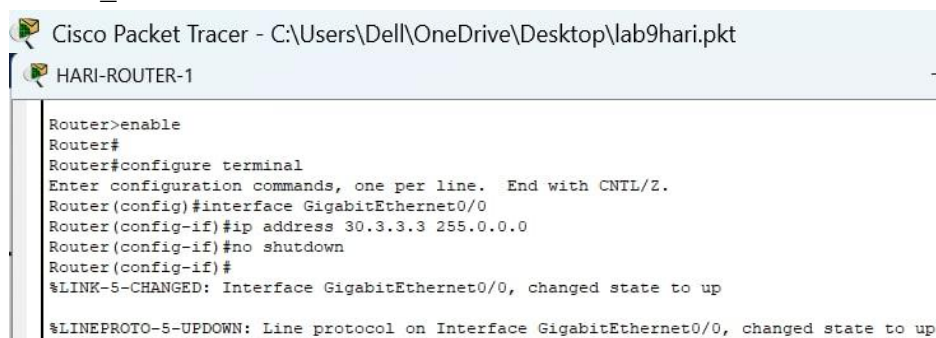
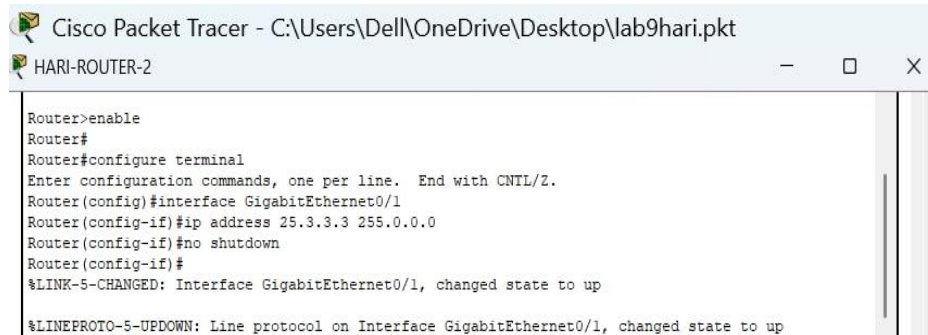


Fig: Configuring IP Address on Gigabit Ethernet Interface

For HARI_ROUTER-2:

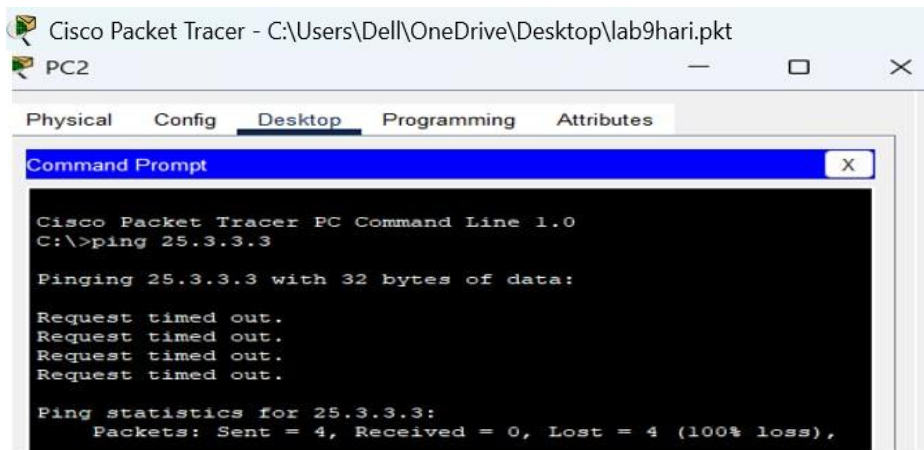


```
Cisco Packet Tracer - C:\Users\Dell\OneDrive\Desktop\lab9hari.pkt
HARI-ROUTER-2

Router>enable
Router#
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface GigabitEthernet0/1
Router(config-if)#ip address 25.3.3.3 255.0.0.0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
```

Fig: Configuring IP Address on Gigabit Ethernet Interface

Step 2: Checking Router is working or not



```
Cisco Packet Tracer - C:\Users\Dell\OneDrive\Desktop\lab9hari.pkt
PC2

Physical Config Desktop Programming Attributes

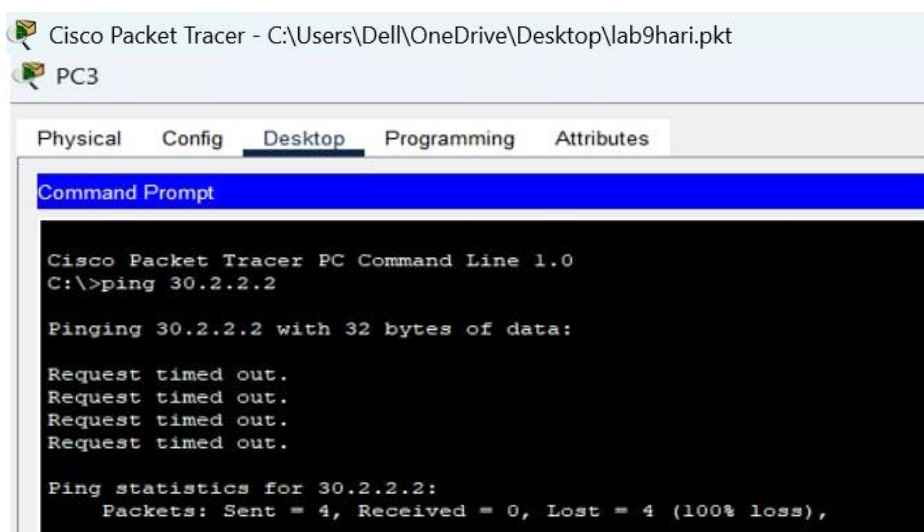
Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 25.3.3.3

Pinging 25.3.3.3 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 25.3.3.3:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```



```
Cisco Packet Tracer - C:\Users\Dell\OneDrive\Desktop\lab9hari.pkt
PC3

Physical Config Desktop Programming Attributes

Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 30.2.2.2

Pinging 30.2.2.2 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

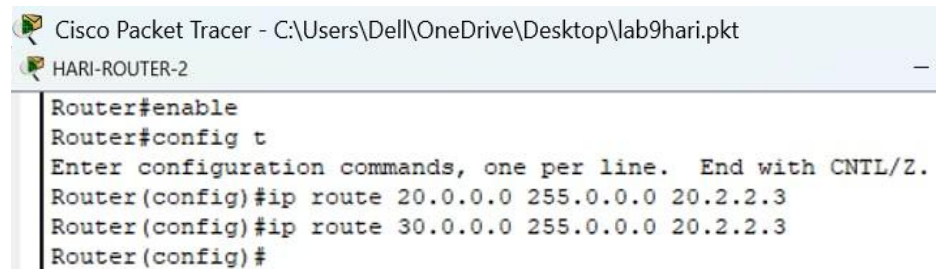
Ping statistics for 30.2.2.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

Fig: Router Connectivity Test Failure

Above figure shows two failed ping attempts from different PCs to check connectivity. Both PCs are trying to reach different Ips, but the destination hosts are unreachable. The error message 100% packet loss indicate a problem with the router's configuration or connection between networks. This suggests that the router is not routing packets correctly between these networks.

Step 3: Adding a static route

Adding a static route to Router 2's network

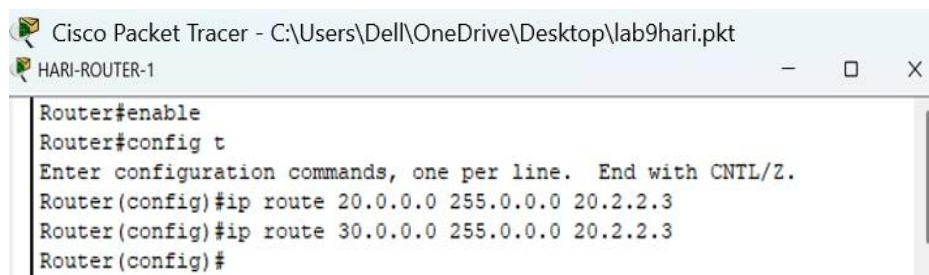


```

Cisco Packet Tracer - C:\Users\Dell\OneDrive\Desktop\lab9hari.pkt
HARI-ROUTER-2
Router#enable
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip route 20.0.0.0 255.0.0.0 20.2.2.3
Router(config)#ip route 30.0.0.0 255.0.0.0 20.2.2.3
Router(config)#
  
```

Fig: Adding Static Route to HARI_ROUTER-2 Network

Add a static route to Router R1's network



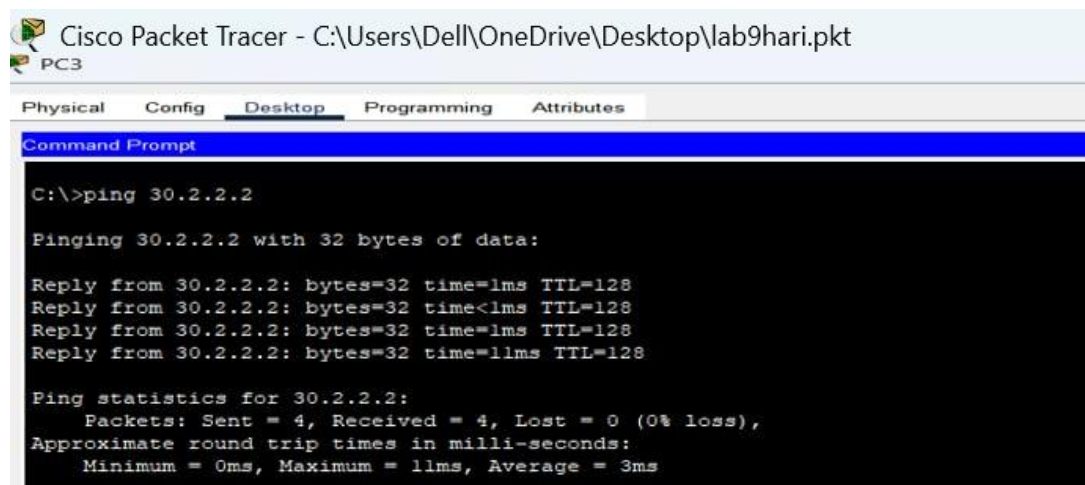
```

Cisco Packet Tracer - C:\Users\Dell\OneDrive\Desktop\lab9hari.pkt
HARI-ROUTER-1
Router#enable
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip route 20.0.0.0 255.0.0.0 20.2.2.3
Router(config)#ip route 30.0.0.0 255.0.0.0 20.2.2.3
Router(config)#
  
```

Fig: Adding Static Route to HARI_ROUTER-1 Network

II. Testing and Validation

From each PC, use the ping command to test connectivity to a PC in a different network.



```

Cisco Packet Tracer - C:\Users\Dell\OneDrive\Desktop\lab9hari.pkt
PC3
Physical Config Desktop Programming Attributes
Command Prompt
C:\>ping 30.2.2.2

Pinging 30.2.2.2 with 32 bytes of data:

Reply from 30.2.2.2: bytes=32 time=1ms TTL=128
Reply from 30.2.2.2: bytes=32 time<1ms TTL=128
Reply from 30.2.2.2: bytes=32 time=1ms TTL=128
Reply from 30.2.2.2: bytes=32 time=11ms TTL=128

Ping statistics for 30.2.2.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 11ms, Average = 3ms
  
```

The screenshot shows the Cisco Packet Tracer interface for PC2. The 'Desktop' tab is selected, displaying a 'Command Prompt' window. The command prompt shows the execution of the 'ping 25.3.3.3' command. The output indicates that the ping was successful, with four replies received from 25.3.3.3, each with 32 bytes of data. The round trip times are shown as <1ms, 13ms, and 3ms. The statistics summary shows 4 packets sent, 4 received, and 0% loss.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 25.3.3.3

Pinging 25.3.3.3 with 32 bytes of data:

Reply from 25.3.3.3: bytes=32 time<1ms TTL=128
Reply from 25.3.3.3: bytes=32 time<1ms TTL=128
Reply from 25.3.3.3: bytes=32 time<1ms TTL=128
Reply from 25.3.3.3: bytes=32 time=13ms TTL=128

Ping statistics for 25.3.3.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 13ms, Average = 3ms
```

Fig: Testing Connectivity Between PC's Using Ping Commands

The above shows successful ping tests between two PCs in different networks. PC2 pings 25.3.3.3 with no packet loss and low latency, while PC3 pings 30.2.2.2 with similar successful results. This confirms network connectivity between the PCs.

Addressing Table:

The addressing table of the static routing is as follows:

Device	Interface	IPv4 Address	Subnet	Default Gateway
PC2	NIC	30.2.2.2	255.0.0.0	30.3.3.3
PC3	NIC	25.3.3.3	255.0.0.0	25.3.3.3

CONCLUSION

In this lab, we explored basic router configuration and static routing using Cisco Packet Tracer. Routers and static routes were set up to enable communication between PCs on different networks, demonstrating essential networking skills and the role of routing in connectivity.